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Carbonate Marine Cementation Based on Flume Tank Experiments

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Abstract

Carbonate diagenesis is one of the key processes that ultimately controls reservoir quality and storage capacity of carbonate reservoirs. The rates of early marine diagenetic reactions in the geological past are little constrained. Fundamental physical experiments allow the calibration of early carbonate diagenesis for a variety of further applications, e.g., process-based modeling, fluid-flow and ultimately, reservoir and storage simulation.

We conducted a year-long flume tank experiment to study cementation along a supratidal to subtidal depositional profile, with the aim of improving our understanding of early marine cementation. The experiment used artificial sea water with periodic oscillation in water level to create supratidal, intertidal and subtidal diagenetic environments. To simulate an active deposition zone in the subtidal environment, we established a "subtidal zone with deposition" by periodically adding thin layers of carbonate sediment on a monthly basis. Monthly sampling was conducted, with four samples collected to characterize the distinct depositional environments. Furthermore, a quantitative diagenetic model was developed to predict the cementation process within the tank, which was calibrated using experimental water chemistry data. This integrated approach allowed for a comprehensive understanding of the depositional and diagenetic processes occurring in the simulated subtidal environment.

The petrographically observed quantities of aragonite cement formed over the period of one year range from ~0-4% of total sediment (grain + porosity) with an estimated average cement fraction of ~1.0%. The calculated cementation rate is 1.96×10^{-9} mol/m²/s, which is comparable with the aragonite growth rate measured in a recent mixed flow reactor experiment in the literature. The cementation rate can be used to calibrate integrated deposition and diagenetic models for modeling temporal and spatial evolution of early carbonate cement at reservoir scale. We estimated that an average cement fraction of ~10% would be generated over 1440 effective years, leading to a substantial reduction in porosity and permeability.

To the best of our knowledge, this study represents the first attempt to physically model early marine diagenetic processes in a flume tank. The approach allows us to estimate cementation effects on the porosity

of carbonate sediment and to predict the yearly amount of carbon mineralization at different atmospheric CO₂ levels.

Introduction

Carbonate diagenesis is a critical process that significantly influences the final quality of carbonate reservoirs. It plays a key role in altering porosity and permeability, which in turn affects reservoir volume, flow properties, and ultimately the commercial viability of the reservoir (Kjensvik et al. 1994; Willis & White 2000; Eaton 2006; Falivene et al. 2006). Among the various diagenetic processes, carbonate cementation stands out as particularly important. Cements fill pore spaces, reducing both pore size and volume, which often leads to a decrease in hydrocarbon storage capacity and production from carbonate reservoirs. However, early cements can also play a beneficial role by preserving the original porosity. They create a solid rock framework at shallow burial depths, thereby mitigating the negative effects of subsequent compaction or additional cementation (Longman 1980; Moore and Wade 2013). Furthermore, if these cements are abundant enough to form multiple layers of submarine lithification (or hardgrounds), they can contribute to reservoir compartmentalization.

Flume tank experiments are widely utilized to study the distribution of sedimentary facies, stratigraphy, and architecture in both carbonate and siliciclastic sediments within specific depositional environments. These experiments offer the advantage of recreating scaled-down depositional environments, allowing for the simulation of hydrodynamic behaviors and depositional processes under controlled laboratory conditions. However, their application remains limited. To the best of the authors' knowledge, no comparable studies have been identified in the existing literature. To address this gap, we conducted a flume tank experiment over a period of one year to investigate the coupled depositional and CaCO₃ cementation processes in a carbonate ramp setting.

Method

Flume tank experiment was conducted at the St. Anthony Falls Laboratory, University of Minnesota. This experiment used the real carbonate sediments as the starting material and synthetic modern seawater to mimic the beach rock formation process, in four depositional environments: supratidal, intertidal, subtidal with sediment growth and without growth (Figure 1). A flume tank with dimensions of 7.3 × 0.9 × 0.1 m was constructed (Figures 2-4). Oolitic aragonite sand from the Bahamas was used. The sediments were dry-sieved to ensure the consistency in grain size.

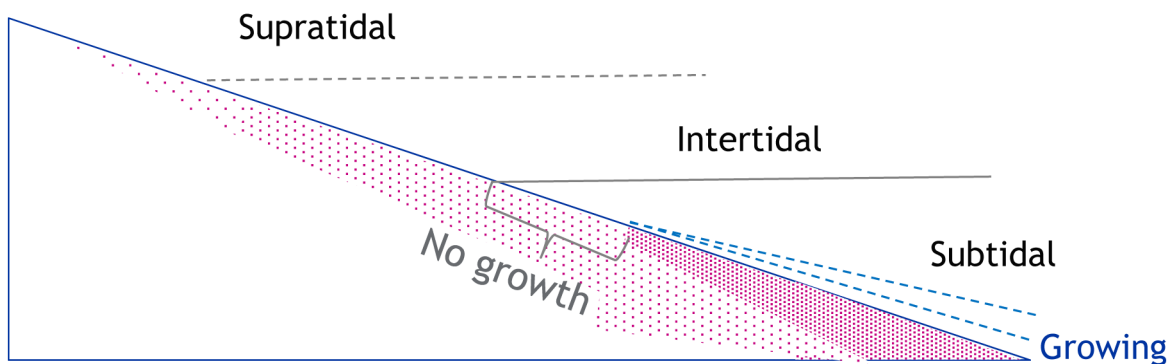


Figure 1—Conceptual model of early marine carbonate cementation in a ramp setting. Due to the fluctuation of seawater, the ramp is divided into supratidal, intertidal and subtidal zones. In the subtidal zone, we further considered the no-growth and growing sub-zones.

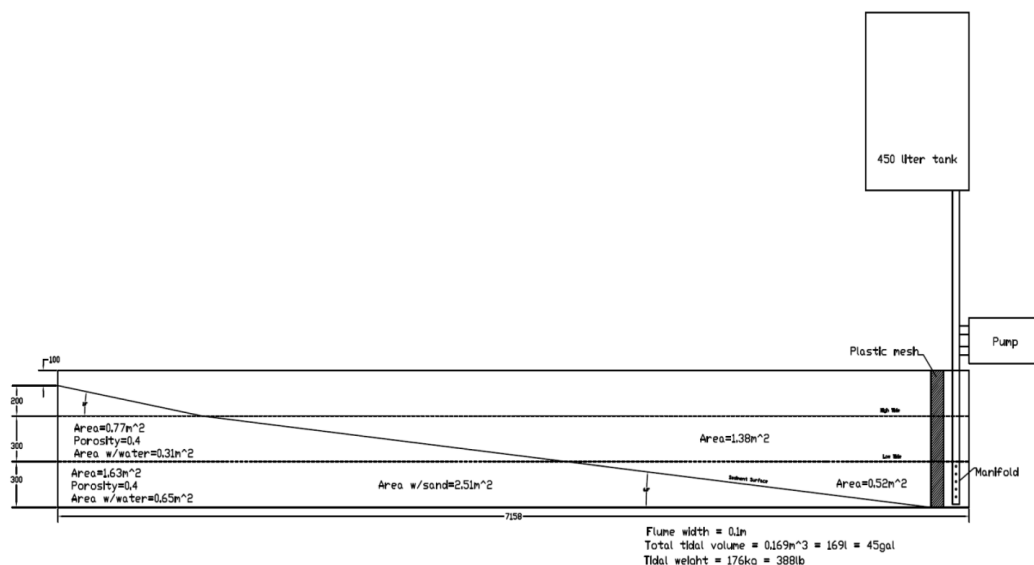


Figure 2—Design of the flume tank according to the conceptual model.



Figure 3—Side view of flume after completed construction.



Figure 4—The flume tank in operation with sand and synthetic seawater.

Synthetic seawater was made and its composition was verified (Table 1). Carbonate cementation consumes Ca^{2+} and HCO_3^- from the water. Within one day or two, the water's capacity of further cementation would be exhausted. Therefore, at end of each working day, 19.17 grams of anhydrous sodium carbonate (Na_2CO_3) and 18.24 grams of calcium chloride (CaCl_2) were added to the reservoir tank in order to compensate for the Ca and C consumption during the CaCO_3 cement formation. The CO_2 concentration in the air in the flume was monitored and recorded daily.

Table 1—Chemistry of synthetic seawater.

Component	Unit (ppm)
Ca	664
Cl	31,011
K	665
Mg	2130
Na	17,885
SO ₄	4321
HCO ₃	150.5
pH	8.0
TDS	56,826

During and after the experiment, water and core samples will be taken at different localities (shoreward to basinward) and different depths. The coring method was based on piston style coring. Two types of cores were acquired. The small cores were taken using clear acrylic tubes with the inner diameter of 16 mm. The coring schedule is shown in Table 2. An example set of small cores is shown in Figure 5. In addition to the small cores, 12 larger cores with the size of 70 mm diameter were taken at the end of the experiments (Table 3). For large cores, core liner and sediment are split length wise into 2 equal halves. One half is designated the archive half and used for non-destructive analysis only. The other half is the working half and is available for sampling.

Table 2—Schematic demonstration for the design of the coring schedule and locations.

Core Date	Core Region	X Coordinate	Y Coordinate
Month 2	Above High Tide	6050	80
	Between Tides	4040	80
	Below Low Tide	2685	80
	Growth Area	1600	80
Month 3	Above High Tide	6050	30
	Between Tides	4040	30
	Below Low Tide	2685	30
	Growth Area	1600	30
Month 4	Above High Tide	6090	70
	Between Tides	4080	70
	Below Low Tide	2725	70
	Growth Area	1640	70
Month 5	Above High Tide	6130	30
	Between Tides	4080	30
	Below Low Tide	2725	30
	Growth Area	1640	30
Month 6	Above High Tide	6130	70
	Between Tides	4120	70
	Below Low Tide	2765	70

<u>Core Date</u>	<u>Core Region</u>	<u>X Coordinate</u>	<u>Y Coordinate</u>
	Growth Area	1680	70
	Above High Tide	6170	30
	Between Tides	4120	30
	Below Low Tide	2765	30
	Growth Area	1680	30
	Above High Tide	6170	70
	Between Tides	4160	70
	Below Low Tide	2600	70
	Growth Area	1560	70
	Above High Tide	6230	30
	Between Tides	4250	30
	Below Low Tide	2570	30
	Growth Area	1530	30
	Above High Tide	6260	70
	Between Tides	4280	70
	Below Low Tide	2540	70
	Growth Area	1500	70
	Above High Tide	6290	30
	Between Tides	4310	30
	Below Low Tide	2510	30
	Growth Area	1470	30
	Above High Tide	6320	70
	Between Tides	4340	70
	Below Low Tide	2825	70
	Growth Area	1440	70

Table 3—Core schedule of 7 cm large cores.

7cm cores		
<u>X Coordinate</u>	<u>Y Coordinate</u>	<u>Tidal Region</u>
6700	50	Supratidal
6400	50	Supratidal
5750	50	Intertidal
4700	50	Intertidal
3700	50	Intertidal
3400	50	Intertidal
3000	50	Subtidal
2300	50	Subtidal
2000	50	Subtidal
1350	50	Growth
1150	50	Growth
950	50	Growth



Figure 5—An example set of core samples taken from the flume tank (Month 6).

Measurements and analysis were performed through different multi scale imaging technology, including high resolution core imaging using Geotek MSCL-CIS core imaging system and scanning electron Microscope (SEM) with energy dispersive X-ray spectroscopy (EDS) analysis. The MSCL-CIS utilizes high performance Geoscan VI camera (with a pixel size of 50 microns). One half of the large core is cleaned and imaged with the high resolution imager at the National Lacustrine Core Facility, University of Minnesota.

Diagenetic modeling

We conducted a diagenetic modeling of aragonite precipitation with the code PhreeqcI v3.4 (Parkhurst and Appelo, 2013). Modern ooids from the Bahamas and the cements are likely to be aragonite, a polymorph of calcium carbonate. The model considered a well-mixed batch experiment open to the atmosphere ($p\text{CO}_2 = -3.382$) with the salt addition every day. Mass conservation is applied in this model. However, this model can only predict the amount of the aragonite precipitated, but not the spatial distribution of the cements.

The modeling results are shown in Figure 6. The experimental data (open diamonds) agree quite well with the numerical model (blue lines). Although the analytical data are somewhat scattered, the main trend is correct and obvious. The model can make predictions for the future evolution of the water chemistry. It was first constructed at 153 days of the experiment. The trend of the samples after the model construction (184, 217, and 244 days) have all been well predicted. In fact, the model has been simulated to the end of the experiment (365 days), which can be further tested and verified in the future samples.

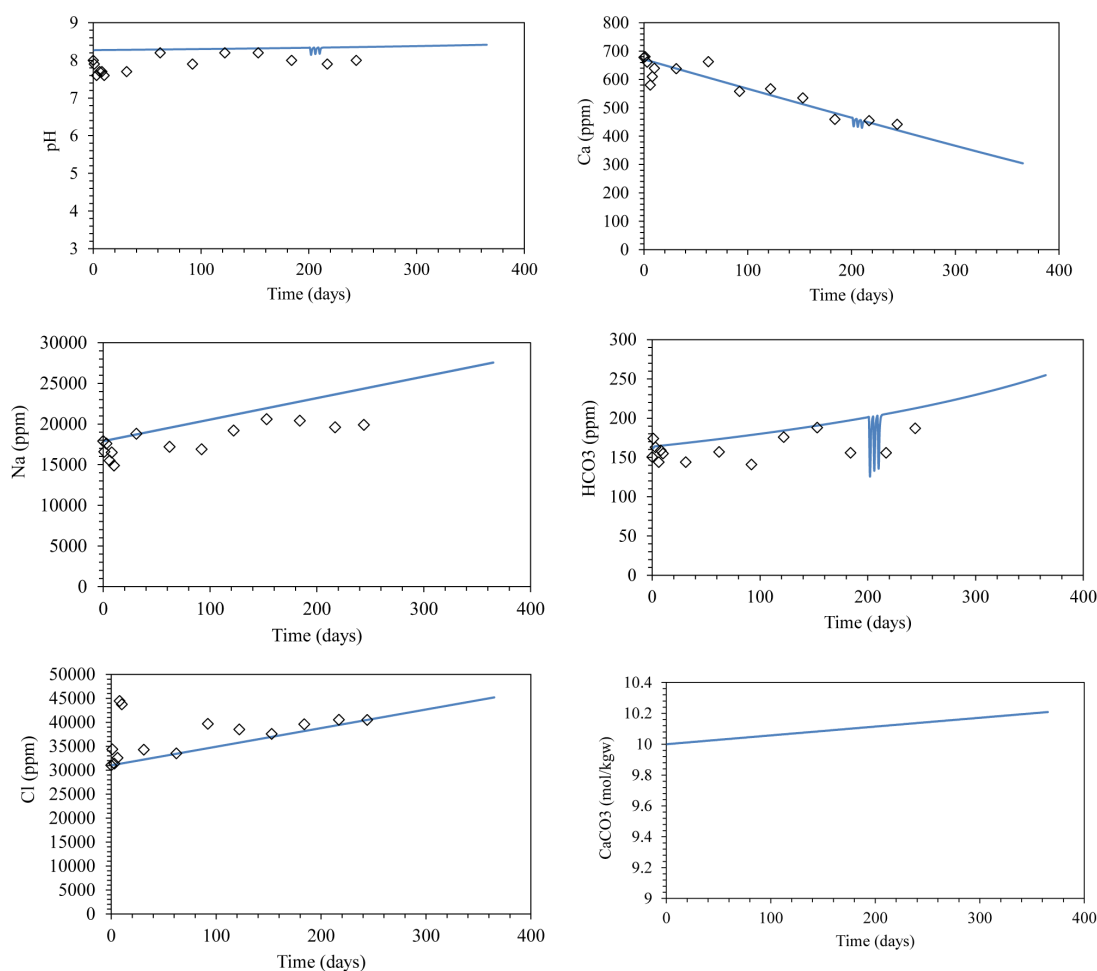


Figure 6—pH and concentrations of Ca, Na, HCO₃ and Cl in ppm as a function of time. Open diamonds are analytical results from the water samples and the blue lines are the results from the numerical model. Some oscillations at about 200 days in the modeling result are due to the convergence problem of the numerical model and can be ignored.

There are some dilution-related issues in the analysis of chloride concentrations in some early samples. For example, the chloride concentrations at 8 and 10 days are apparently outliers, so are the 31 and 62 days results. However, the re-measured values for 31 and 62 days results pulled them back to the normal trend.

The decrease of Ca concentration is due to the consumption of aragonite precipitation, while the HCO₃ still increases (like Na and Cl) due to the compensation of CO₂ from the atmosphere. Because the saturation index for aragonite (CaCO₃) will be $[Ca^{+2}] \times [CO_3^{-2}]$, the value will remain almost constant (Ca⁺² decreasing while CO₃⁻² increasing). Therefore, the cementation process will keep going.

The model is capable of predicting the total amount of aragonite precipitated, but it does not provide information on the spatial distribution of the cement. The model predicted aragonite growth rate, based on the total precipitated aragonite (~ 0.2 mol/kgw for 365 d), is approximately 6.64×10^{-9} mol/kgw/s. To translate this rate into a more comparable form, we considered a geometric surface area of 53 cm²/g for 0.4 mm aragonite grains (Eq. 1), assuming 640 g of aragonite in 1 kg of water. This results in an aragonite cementation rate of 1.96×10^{-9} mol/m²/s. This rate is within the range of measured aragonite growth rates from the literature, which accounts for magnesium incorporation (2.51×10^{-9} to 7.94×10^{-8} mol/m²/s; Mavromatis et al., 2022). This comparison underscores the relevance of our findings to existing geochemical studies on aragonite cementation.

The external specific surface area SA (m²/g) of a smooth particle is that particle's area divided by the volume times the density of the material for spherical particles:

$$SA = SA = \frac{4\pi R^2}{\frac{4}{3}\rho\pi R^3} = \frac{3}{\rho R} \quad (1)$$

The initial porosity of the sediment was approximately 30%, as confirmed by computerized tomography (CT) analysis. Cementation was found to have a relatively small impact on porosity. Following the precipitation of 0.21 mol/kgw of aragonite, the calculated final porosity was estimated to be around 29%, assuming a uniform distribution of the aragonite cement.

Unlike conventional methods, which typically employ batch-type and mixed flow reactors (e.g., [Lu et al., 2013](#)) to determine mineral dissolution or precipitation rates, the rate obtained in this study more closely replicates the depositional and environmental conditions present during aragonite precipitation. As a key parameter in integrated deposition and diagenetic models, the aragonite cement growth rate has significant implications for modeling early marine carbonate cementation. The rate determined in this study can be used to calibrate these models, enabling more accurate predictions of the temporal and spatial distributions of cementation at the reservoir scale, as demonstrated in recent studies (e.g., [Lu et al., 2023](#)).

Petrographic analysis

Petrographic analysis was performed to identify cement deposition in the flume tank. High resolution core images identified the possible calcite precipitation (isopaches blades) and SEM analysis found spherulitic fibrous aragonite ([Figures 7 and 8](#)). SEM-EDS analysis on random samples collected at all zones shows that CaCO_3 cement (mainly Aragonite) was found in all the tidal zones from the slabbed whole cores. The cement was not evenly distributed throughout the samples where it was observed. The quantity of the cement could not be determined from imaging due to the small quantity.



Figure 7—Possible calcite precipitation (isopaches blades; pointed by the black arrows) in the pore spaces connecting the oolitic grains (Subtidal Sample, x 2300). The location of the sample in the large core is shown on the right.

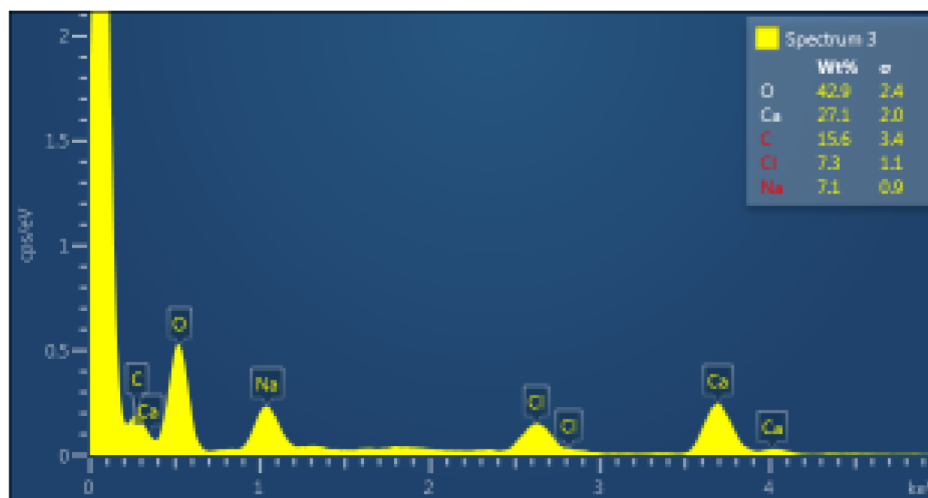
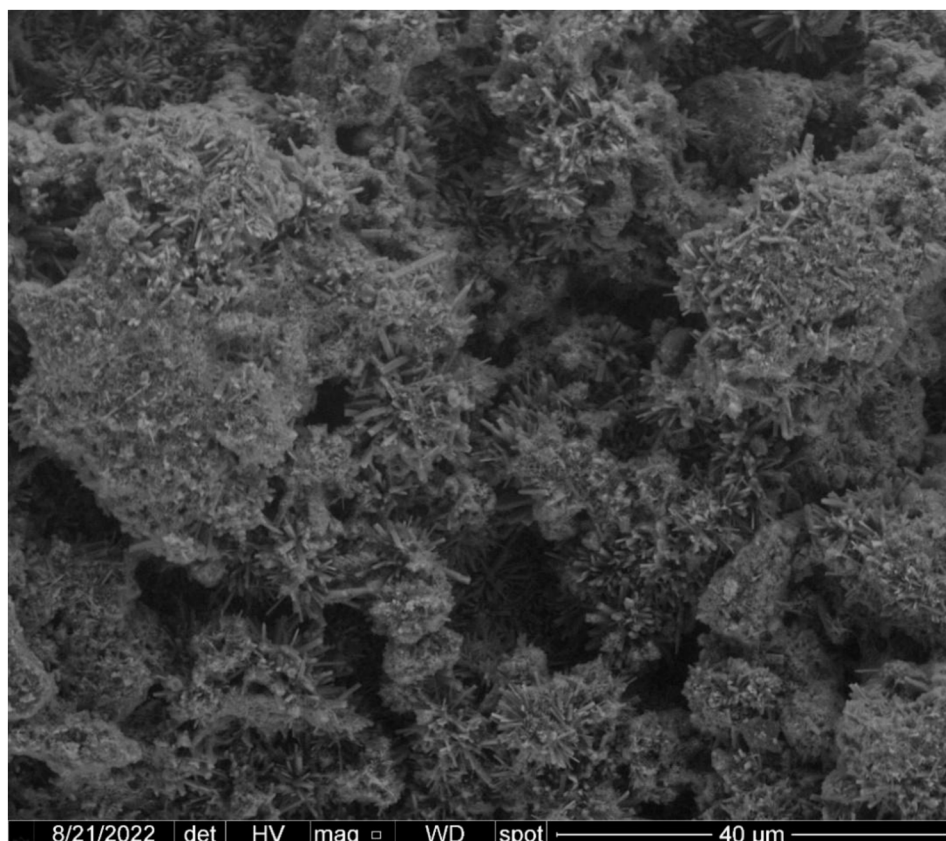


Figure 8—SEM Images for growth of spherulitic fibrous aragonite (whisker crystals) as bridges between ooid (Sample X1150 –Growth Zone-grains). EDS analysis confirmed the presence of aragonite (CaCO_3).

While a precise quantification of cement abundance in each zone was not feasible, estimates based on high-resolution core photos and SEM images suggest that cement quantities typically range from approximately 0-4% for 1440 effective years of semi-diurnal tide. For comparison, Flügel and Koch (1995) estimated the abundance of marine cement for the slope setting ranges from 0 - 4% and is generally less than 7% for the Late Triassic Steinplatte carbonate platform (Austria). Wasserman (2011) summarized the cement abundance for ten different reefs and the cement abundance ranges from <1% to 11.6%. Lu et al. (2023) conducted modeling with a dynamic depositional framework and reactive transport modeling for

carbonate cementation in a relict shelf-edge coral reef North of Miami, Florida. They predicted about 5.1% cement growth for 4,000 yrs.

Conclusions

A one-year flume tank experiment was designed to simulate early marine cementation in a synthetic carbonate ramp setting, comprising aragonitic oolitic sands and four distinct sub-environments. This laboratory-based study successfully replicated the process of aragonite cement generation in a tidal depositional system. The resulting cements were characterized by three primary types: fibrous aragonite crystals, nano-sized spherical calcite grains, and euhedral halite plates, with the latter likely forming through the desiccation of dissolved salts. Notably, cement formation exhibited a decreasing trend in abundance from subtidal to supratidal environments, with quantities ranging from approximately 0% to 4%. Diagenetic modeling of the experiment revealed an average cement generation of around 1% and a corresponding porosity reduction of approximately 1%. Furthermore, the calculated aragonite growth rate of 1.96×10^{-9} mol/m²/s provides a valuable parameter for calibrating integrated depositional-diagenetic models at the reservoir scale, enabling more accurate forecasting of carbonate reservoir performance.

The experimental work conducted in this study provides insights into the time required for measurable changes in the porosity, permeability, and/or erodibility of carbonate sediment within a shoreline depositional system. Assuming that an average of ~10% cement is sufficient to impact the reservoir and storage quality of carbonate reservoirs, the calculation indicated that at the current experimental conditions approximately 10 years would be needed to achieve ~10% aragonite cementation. This duration is equivalent to 1440 effective years of semi-diurnal tide.

Future research directions could involve exploring the effects of different carbonate sediments, such as fresh reef fragments or algae mound fragments, on the system. Additionally, increasing heterogeneity by mixing carbonate sand and mud could provide valuable insights. Further experiments could also investigate the impact of elevated salinity levels (e.g., 2-3 times that of seawater), higher temperatures (e.g., 30°C, achieved through constant water heating), and extended experimental durations (e.g., 3 years). This would allow for a more comprehensive understanding of the system's behavior under various conditions.

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